Solution

With usual notations in triangle ABC, we have $\overrightarrow{BC}=\vec{a}$, $\overrightarrow{CA}=\vec{b}$, and $\overrightarrow{AB}=\vec{c}$. Then

$|\overrightarrow{BC}|=a$, $|\overrightarrow{CA}|=b$, $|\overrightarrow{AB}|=c$ and $\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB}=\vec{0}$

So, $\overrightarrow{BC}=-\overrightarrow{CA}-\overrightarrow{AB}$

Applying dot product, we get

$$\overrightarrow{BC}\cdot\overrightarrow{BC} = -\overrightarrow{BC}\cdot\overrightarrow{CA}-\overrightarrow{BC}\cdot\overrightarrow{AB}$$

$$\Rightarrow\ |\overrightarrow{BC}|^2 = -|\overrightarrow{BC}||\overrightarrow{CA}|\cos(\pi-C)-|\overrightarrow{BC}||\overrightarrow{AB}|\cos(\pi-B)$$

$$\Rightarrow\ a^2 = ab\cos C+ac\cos B$$

Therefore $a=b\cos C+c\cos B$. The results in (ii) and (iii) are proved in a similar way.

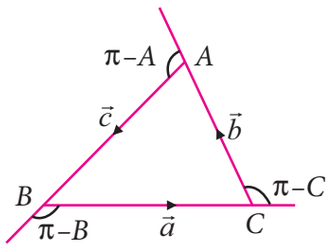


Fig. 6.7

Example 6.3

By vector method, prove that $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$.

Solution

Let $\hat{a}=\overrightarrow{OA}$ and $\hat{b}=\overrightarrow{OB}$ be the unit vectors and which make angles α and β, respectively, with positive x-axis, where A and B are as in the Fig. 6.8. Draw AL and BM perpendicular to the x-axis. Then $|\overrightarrow{OL}|=|\overrightarrow{OA}|\cos\alpha=\cos\alpha$, $|\overrightarrow{LA}|=|\overrightarrow{OA}|\sin\alpha=\sin\alpha$.

So, $\overrightarrow{OL}=|\overrightarrow{OL}|\,\hat{i}=\cos\alpha\hat{i}$, $\overrightarrow{LA}=\sin\alpha(-\hat{j})$.

Therefore, $\hat{a}=\overrightarrow{OA}=\overrightarrow{OL}+\overrightarrow{LA}=\cos\alpha\hat{i}-\sin\alpha\hat{j}$. ... (1)

Similarly, $\hat{b}=\cos\beta\hat{i}+\sin\beta\hat{j}$ ... (2)

The angle between $\hat{a}$ and $\hat{b}$ is $\alpha+\beta$ and so,

$$\hat{a}\cdot\hat{b}=|\hat{a}||\hat{b}|\cos(\alpha+\beta)=\cos(\alpha+\beta) \quad \text{... (3)}$$

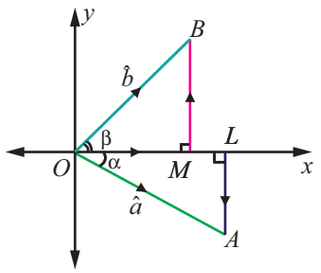


Fig. 6.8

On the other hand, from (1) and (2)

$$\hat{a}\cdot\hat{b}=(\cos\alpha\hat{i}-\sin\alpha\hat{j})\cdot(\cos\beta\hat{i}+\sin\beta\hat{j})=\cos\alpha\cos\beta-\sin\alpha\sin\beta. \quad \text{... (4)}$$

From (3) and (4), we get $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$.

Example 6.4

With usual notations, in any triangle ABC, prove by vector method that $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$.

Solution

With usual notations in triangle ABC, we have $\overrightarrow{BC}=\vec{a}$, $\overrightarrow{CA}=\vec{b}$, and $\overrightarrow{AB}=\vec{c}$. Then $|\overrightarrow{BC}|=a$, $|\overrightarrow{CA}|=b$, and $|\overrightarrow{AB}|=c$.

Since in ΔABC, $\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB}=0$, we have $\overrightarrow{BC}\times(\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB})=\vec{0}$.

Simplification gives,

$$\overrightarrow{BC}\times\overrightarrow{CA}=\overrightarrow{AB}\times\overrightarrow{BC}. \quad \text{... (1)}$$

Similarly, since $\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB}=\vec{0}$, we have

$$\overrightarrow{CA}\times(\overrightarrow{BC}+\overrightarrow{CA}+\overrightarrow{AB})=\vec{0}.$$

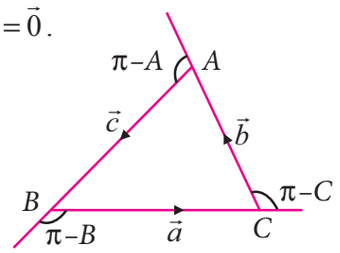


Fig. 6.9

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